EXTRA LAB EXERCISES:

Lab-1 : Create a database called library_db and a table books with columns: book_id, title, author, publisher, year_of_publication, and price. Insert five records into the table.

Query :-

Step 1 :- creating a Database

CREATE DATABASE library_db;

Step 2:- Creating a table of books

CREATE TABLE books(book_id int, title varchar(100), author varchar(100), publisher varchar(100), year_of_publication int, price int);

Step 3 :- Insert a five records in books

INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('The Alchemist', 'Paulo Coelho', 'HarperCollins', 1988, 399);

INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('To Kill a Mockingbird', 'Harper Lee', 'J.B. Lippincott & Co.', 1960, 499);

INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('1984', 'George Orwell', 'Secker & Warburg', 1949, 350);

INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('Pride and Prejudice', 'Jane Austen', 'T. Egerton', 1813, 299);

INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('The Great Gatsby', 'F. Scott Fitzgerald', 'Charles Scribner\'s Sons', 1925, 450);

SCREENSHOT :-

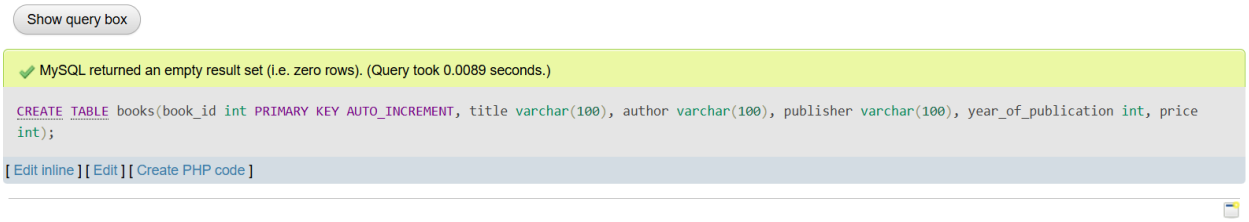


Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	book_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	title	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	author	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	publisher	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More
5	year_of_publication	int(11)			Yes	NULL			Change Drop More
6	price	int(11)			Yes	NULL			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Spatial Fulltext

1 row inserted.
Inserted row id: 1 (Query took 0.0023 seconds.)

```
INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('The Alchemist', 'Paulo Coelho', 'HarperCollins', 1988, 399);
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 2 (Query took 0.0014 seconds.)

```
INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('To Kill a Mockingbird', 'Harper Lee', 'J.B. Lippincott & Co.', 1960, 499);
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 3 (Query took 0.0013 seconds.)

```
INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('1984', 'George Orwell', 'Secker & Warburg', 1949, 350);
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 4 (Query took 0.0016 seconds.)

```
INSERT INTO books(title, author, publisher, year_of_publication, price)VALUES('Pride and Prejudice', 'Jane Austen', 'T. Egerton', 1813, 299);
```

Console

Extra options

	book_id	title	author	publisher	year_of_publication	price
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	399
Edit Copy Delete	2	To Kill a Mockingbird	Harper Lee	J.B. Lippincott & Co.	1960	499
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	350
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	1813	299
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	450

Check all *With selected:* Edit Copy Delete Export

Lab-2 : Create a table members in library_db with columns: member_id, member_name, date_of_membership, and email. Insert five records into this table.

Query :-

Step 1 :- Creating a table for members in library_db database

CREATE TABLE members(member_id int Primary key, member_name varchar(100), date_of_membership DATE, email varchar(100));

Step 2 :- Insert five records in member table

INSERT INTO members(member_name,date_of_membership,email)VALUES('Yash Tailor', '2026-01-15', 'yash.tailor@gmail.com');

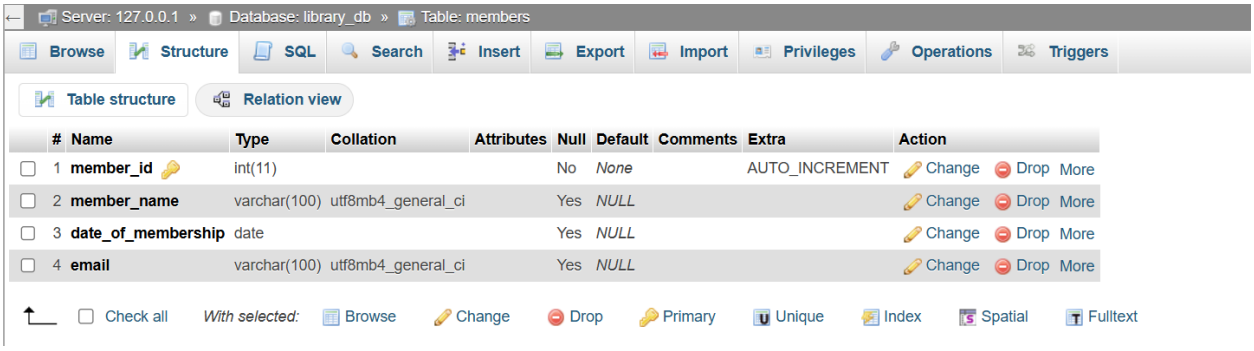
INSERT INTO members(member_name,date_of_membership,email)VALUES('Harsh Lakdiwala', '2026-03-22', 'harsu.lakdiwala@gmail.com');

INSERT INTO members(member_name,date_of_membership,email)VALUES('Parth Gamit', '2023-05-10', 'parth.gamit@gmail.com');

INSERT INTO members(member_name,date_of_membership,email)VALUES('Yash Gandhi', '2023-07-18', 'yash.gandhi@gmail.com');

INSERT INTO members(member_name,date_of_membership,email)VALUES('Ronak Patel', '2023-09-05', 'Ronak.patel@gmail.com');

SCREENSHOT :-



member_id	member_name	date_of_membership	email
1	Yash Tailor	2026-01-15	yash.tailor@gmail.com
2	Harsh Lakdiwala	2026-03-22	harsu.lakdiwala@gmail.com
3	Parth Gamit	2023-05-10	parth.gamit@gmail.com
4	Yash Gandhi	2023-07-18	yash.gandhi@gmail.com
5	Ronak Patel	2023-09-05	Ronak.patel@gmail.com
6	Rishi Naik	2020-04-14	rishi.naik@gmail.com
7	Mit Lad	2018-06-12	mit.lad12@gmail.com

Lab-3 : Retrieve all members who joined the library before 2022. Use appropriate SQL syntax with WHERE and ORDER BY.

Query :-

SELECT member_name, date_of_membership FROM members WHERE date_of_membership < ‘2022-01-01’ ORDER BY member_name;

SCREENSHOT:-

Extra options

member_name 1	date_of_membership
Mit Lad	2018-06-12
Rishi Naik	2020-04-14

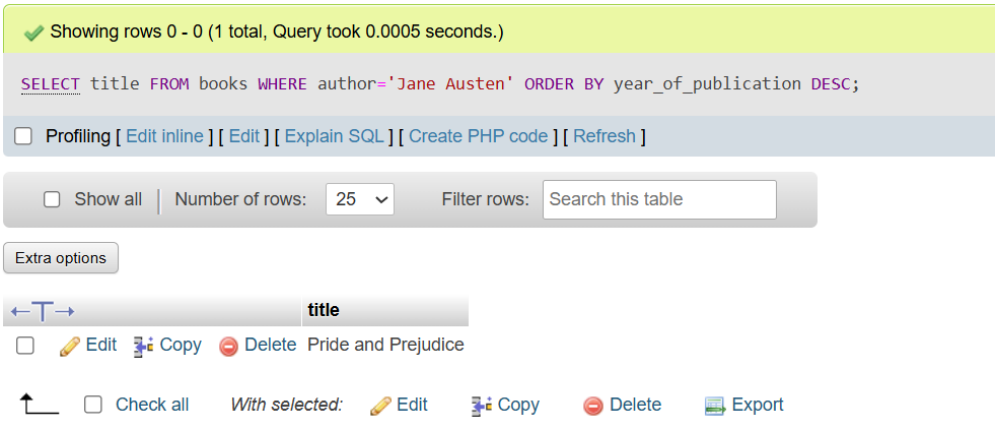
Check all *With selected:* Edit Copy Delete Export

Lab-4 : Write SQL queries to display the titles of books published by a specific author. Sort the results by year_of_publication in descending order.

Query :-

SELECT title FROM books WHERE author='Jane Austen' ORDER BY year_of_publication DESC;

SCREENSHOT:-

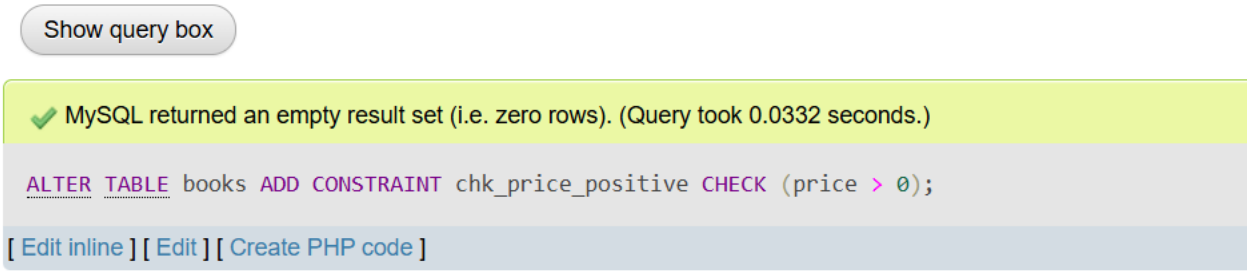


Lab-5 : Add a CHECK constraint to ensure that the price of books in the books table is greater than 0.

Query :-

ALTER TABLE books ADD CONSTRAINT chk_price_positive CHECK (price > 0);

SCREENSHOT:-



Lab-6 : Modify the members table to add a UNIQUE constraint on the email column, ensuring that each member has a unique email address.

Query :-

ALTER TABLE members

ADD CONSTRAINT unique_email

UNIQUE (email);

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0151 seconds.)

```
ALTER TABLE members ADD CONSTRAINT unique_email UNIQUE (email);
```

[Edit inline] [Edit] [Create PHP code]

Lab-7 : Create a table authors with the following columns: author_id, first_name, last_name, and country. Set author_id as the primary key.

Query :-

CREATE TABLE authors(author_id int PRIMARY KEY AUTO_INCREMENT, first_name varchar(100), last_name varchar(100), country varchar(100));

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0090 seconds.)

```
CREATE TABLE authors(author_id int PRIMARY KEY AUTO_INCREMENT, first_name varchar(100), last_name varchar(100), country varchar(100));
```

[Edit inline] [Edit] [Create PHP code]

Lab-8 : Create a table publishers with columns: publisher_id, publisher_name, contact_number, and address. Set publisher_id as the primary key and contact_number as unique.

Query :-

CREATE TABLE publishers(publisher_id int PRIMARY KEY AUTO_INCREMENT, publishers_name varchar(100), contact_number varchar(15) UNIQUE, address varchar(200));

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0289 seconds.)

```
CREATE TABLE publishers(publisher_id int PRIMARY KEY AUTO_INCREMENT, publishers_name varchar(100), contact_number varchar(15) UNIQUE, address varchar(200));
```

[Edit inline] [Edit] [Create PHP code]

Lab-9 : Add a new column genre to the books table. Update the genre for all existing records.

Query :-

Step 1 :- Add New Column from the books table.

ALTER TABLE books ADD genre varchar(50);

Step 2 :- Update the genre for all existing records.

UPDATE books SET genre = ‘Fiction’ WHERE book_id = 1;

UPDATE books SET genre = ‘Classic’ WHERE book_id = 2;

UPDATE books SET genre = ‘Dystopian’ WHERE book_id = 3;

UPDATE books SET genre = ‘Romance’ WHERE book_id = 4;

UPDATE books SET genre = ‘Classic’ WHERE book_id = 5;

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0137 seconds.)

```
ALTER TABLE books ADD genre varchar(50);
```

[Edit inline] [Edit] [Create PHP code]

←T→	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	399	NULL
Edit Copy Delete	2	To Kill a Mockingbird	Harper Lee	J.B. Lippincott & Co.	1960	499	NULL
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	350	NULL
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	1813	299	NULL
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	450	NULL

1 row affected. (Query took 0.0017 seconds.)

```
UPDATE books SET genre = 'Fiction' WHERE book_id = 1;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0012 seconds.)

```
UPDATE books SET genre = 'Classic' WHERE book_id = 2;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0013 seconds.)

```
UPDATE books SET genre = 'Dystopian' WHERE book_id = 3;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0014 seconds.)

```
UPDATE books SET genre = 'Romance' WHERE book_id = 4;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0014 seconds.)

```
UPDATE books SET genre = 'Classic' WHERE book_id = 5;
```

book_id	title	author	publisher	year_of_publication	price	genre
1	The Alchemist	Paulo Coelho	HarperCollins	1988	399	Fiction
2	To Kill a Mockingbird	Harper Lee	J.B. Lippincott & Co.	1960	499	Classic
3	1984	George Orwell	Secker & Warburg	1949	350	Dystopian
4	Pride and Prejudice	Jane Austen	T. Egerton	1813	299	Romance
5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	450	Classic

Check all *With selected:* Edit Copy Delete Export

Lab-10 : Modify the members table to increase the length of the email column to 100 characters.

Query :-

Already column size varchar(100) completed.

Lab-11 : Drop the publishers table from the database after verifying its structure.

Query :-

DROP TABLE publishers;

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0059 seconds.)

```
DROP TABLE publishers;
```

[Edit inline] [Edit] [Create PHP code]

Lab-12 : Create a backup of the members table and then drop the original members table

Query :-

Step 1 :- create a backup member table

CREATE TABLE member_backup AS

SELECT *FROM members;

Step 2 :- verify the member_backup table then DROP the original member table

DROP TABLE members;

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0112 seconds.)

```
CREATE TABLE member_backup AS SELECT *FROM members;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

member_id	member_name	date_of_membership	email
1	Yash Tailor	2026-01-15	yash.tailor@gmail.com
2	Harsh Lakdiwala	2026-03-22	harsu.lakdiwala@gmail.com
3	Parth Gamit	2023-05-10	parth.gamit@gmail.com
4	Yash Gandhi	2023-07-18	yash.gandhi@gmail.com
5	Ronak Patel	2023-09-05	Ronak.patel@gmail.com
6	Rishi Naik	2020-04-14	rishi.naik@gmail.com
7	Mit Lad	2018-06-12	mit.lad12@gmail.com

Show all | Number of rows: 25 Filter rows: Search this table

DROP TABLE member :-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0079 seconds.)

```
DROP TABLE members;
```

[Edit inline] [Edit] [Create PHP code]

Lab-13 : Insert three new authors into the authors table, then update the last name of one of the authors.

Query :-

Step 1 :- Insert new authors into authors table.

INSERT INTO authors(first_name,last_name,country)VALUES('George','Orwell','United Kingdom');

INSERT INTO authors(first_name,last_name,country)VALUES('Jane','Austen','United Kingdom');

INSERT INTO authors(first_name,last_name,country)VALUES('Mark','Twain','United States');

Step 2 :- update the last name of one of the authors.

UPDATE authors SET last_name = 'Taylor' WHERE author_id=2;

SCREENSHOT:-

Insert the Three authors into authors table:-

1 row inserted.
Inserted row id: 1 (Query took 0.0019 seconds.)

```
INSERT INTO authors(first_name,last_name,country)VALUES('George','Orwell','United Kingdom');
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 2 (Query took 0.0020 seconds.)

```
INSERT INTO authors(first_name,last_name,country)VALUES('Jane','Austen','United Kingdom');
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 3 (Query took 0.0013 seconds.)

```
INSERT INTO authors(first_name,last_name,country)VALUES('Mark','Twain','United States');
```

[Edit inline] [Edit] [Create PHP code]

				author_id	first_name	last_name	country
☐	Edit	Copy	Delete	1	George	Orwell	United Kingdom
☐	Edit	Copy	Delete	2	Jane	Austen	United Kingdom
☐	Edit	Copy	Delete	3	Mark	Twain	United States

Check all *With selected:* Edit Copy Delete Export

update the last name of one of the authors :-

				author_id	first_name	last_name	country
☐	Edit	Copy	Delete	1	George	Orwell	United Kingdom
☐	Edit	Copy	Delete	2	Jane	Taylor	United Kingdom
☐	Edit	Copy	Delete	3	Mark	Twain	United States

Check all *With selected:* Edit Copy Delete Export

Lab-14 : Delete a book from the books table where the price is higher than 450.

Query :-

DELETE FROM books WHERE price > 450;

SCREENSHOT:-

1 row deleted. (Query took 0.0009 seconds.)

```
DELETE FROM books WHERE price > 450;
```

[Edit inline] [Edit] [Create PHP code]

←T→	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	399	Fiction
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	350	Dystopian
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	1813	299	Romance
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	450	Classic

Check all *With selected:* Edit Copy Delete Export

Lab-15 : Update the year_of_publication of a book with a specific book_id.

Query :-

UPDATE books SET year_of_publication = 2016 WHERE book_id = 4;

SCREENSHOT:-

1 row affected. (Query took 0.0012 seconds.)

```
UPDATE books SET year_of_publication = 2016 WHERE book_id = 4;
```

[Edit inline] [Edit] [Create PHP code]

←T→	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	399	Fiction
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	350	Dystopian
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	2016	299	Romance
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	450	Classic

Check all *With selected:* Edit Copy Delete Export

Lab-16 : Increase the price of all books published before 2015 by 10%.

Query :-

UPDATE books SET price = price * 1.10 WHERE year_of_publication < 2015;

SCREENSHOT:-

3 rows affected. (Query took 0.0013 seconds.)

```
UPDATE books SET price = price * 1.10 WHERE year_of_publication < 2015;
```

[Edit inline] [Edit] [Create PHP code]

	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	439	Fiction
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	385	Dystopian
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	2016	299	Romance
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	495	Classic

Check all *With selected:* Edit Copy Delete Export

Lab-17 : Remove all members who joined before 2020 from the members table.

Query :-

DELETE FROM member_backup WHERE date_of_membership < '2020-01-01';

SCREENSHOT:-

1 row deleted. (Query took 0.0012 seconds.)

```
DELETE FROM member_backup WHERE date_of_membership < '2020-01-01';
```

[Edit inline] [Edit] [Create PHP code]

member_id	member_name	date_of_membership	email
1	Yash Tailor	2026-01-15	yash.tailor@gmail.com
2	Harsh Lakdiwala	2026-03-22	harsu.lakdiwala@gmail.com
3	Parth Gamit	2023-05-10	parth.gamit@gmail.com
4	Yash Gandhi	2023-07-18	yash.gandhi@gmail.com
5	Ronak Patel	2023-09-05	Ronak.patel@gmail.com
6	Rishi Naik	2020-04-14	rishi.naik@gmail.com

Show all | Number of rows: 25 Filter rows: Search this table

Lab-18 : Delete all books that have a NULL value in the author column

Query :-

DELETE FROM books WHERE author = ‘ ’;

SCREENSHOT:-

	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	439	Fiction
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	385	Dystopian
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	2016	299	Romance
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	495	Classic

Check all *With selected:* Edit Copy Delete Export

Lab-19 : Write a query to retrieve all books with price between 100 and 300.

Query :-

SELECT *FROM books WHERE price BETWEEN 100 AND 300;

SCREENSHOT:-

	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	2016	299	Romance

Check all *With selected:* Edit Copy Delete Export

Lab-20 : Write a query to retrieve all books with price between 100 and 300.

Query :-

SELECT *FROM books ORDER BY author ASC LIMIT 3;

SCREENSHOT:-

	book_id	title	author 1	publisher	year_of_publication	price	genre
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	495	Classic
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	385	Dystopian
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	2016	299	Romance

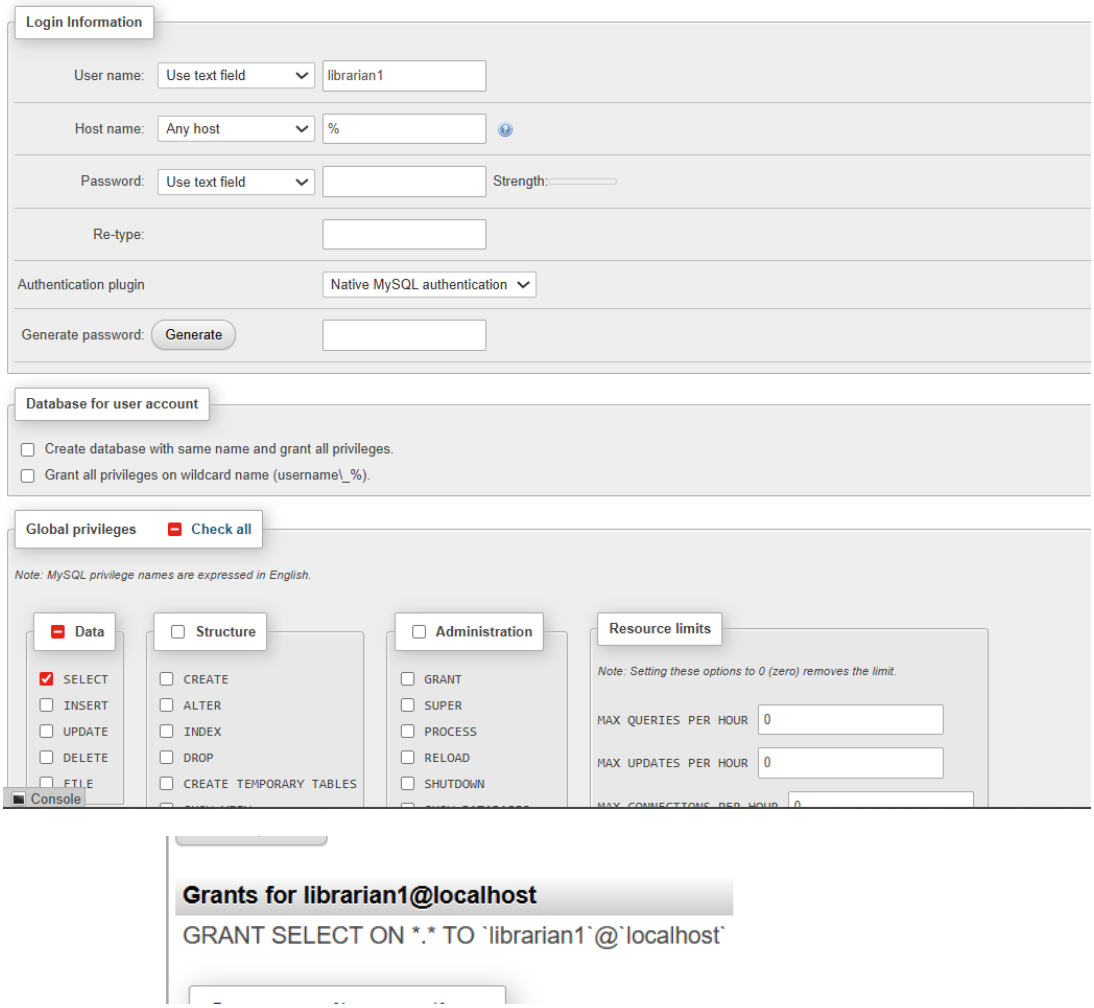
Check all *With selected:* Edit Copy Delete Export

Lab-21 : Grant SELECT permission to a user named librarian on the books table.

Query :-

CREATE USER ‘librarian1’@'localhost' IDENTIFIED BY '123';

SCREENSHOT:-



Lab-22 : Grant INSERT and UPDATE permissions to the user admin on the members table.

Query :-

CREATE USER ‘admin’@'localhost' IDENTIFIED BY '123';

GRANT INSERT, UPDATE ON library_db.members TO 'admin'@'localhost';

FLUSH PRIVILEGES;

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0030 seconds.)

```
CREATE USER 'admin'@'localhost' IDENTIFIED BY '123';
```

[Edit inline] [Edit] [Create PHP code]

Grants for admin@localhost

GRANT SELECT, INSERT, UPDATE ON *.* TO `admin`@`lo...

Insert the Record:-

1 row inserted. (Query took 0.0034 seconds.)

```
INSERT INTO member_backup(member_name,date_of_membership,email)VALUES('Ankit Patel','2026-04-15','ankit123@gmail.com');
```

[Edit inline] [Edit] [Create PHP code]

Update the Record:-

1 row affected. (Query took 0.0062 seconds.)

```
UPDATE member_backup SET member_name = 'Aniket Patel', member_id = 7 WHERE member_id = 0;
```

[Edit inline] [Edit] [Create PHP code]

member_id	member_name	date_of_membership	email
1	Yash Tailor	2026-01-15	yash.tailor@gmail.com
2	Harsh Lakdiwala	2026-03-22	harsu.lakdiwala@gmail.com
3	Parth Gamit	2023-05-10	parth.gamit@gmail.com
4	Yash Gandhi	2023-07-18	yash.gandhi@gmail.com
5	Ronak Patel	2023-09-05	Ronak.patel@gmail.com
6	Rishi Naik	2020-04-14	rishi.naik@gmail.com
7	Aniket Patel	2026-04-15	ankit123@gmail.com

Lab-23 : Revoke the INSERT privilege from the user librarian on the books table.

Query :-

INSERT INTO books(title,author,publisher,year_of_publication,price,genre)VALUES('Tom&Jerry','Warner Brothers','Warner Brothers','2006','250'.'comedy');

SCREENSHOT:-

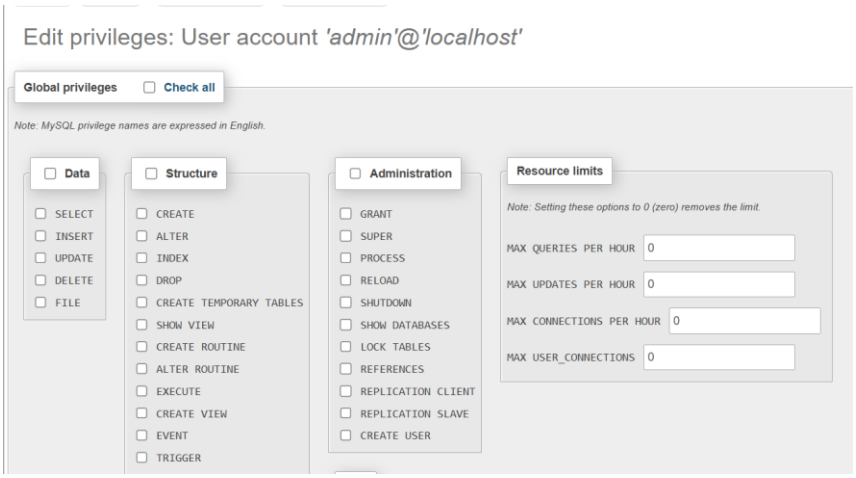
Lab-24 : Revoke all permissions from user admin on the members table.

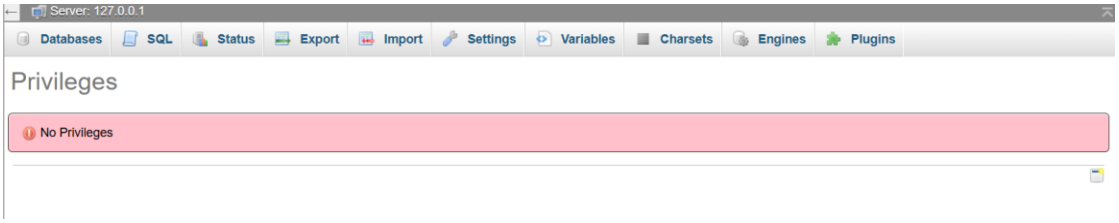
Query :-

REVOKE ALL PRIVILEGES ON library_db.members FROM 'admin'@'localhost';

FLUSH PRIVILEGES;

SCREENSHOT:-





Lab-25 : Use COMMIT after inserting multiple records into the books table, then make another insertion and perform a ROLLBACK.

Query :-

Step 1 :- first insert multiple records and then COMMIT

START TRANSACTION;

INSERT INTO books(title,author,publisher,year_of_publication,price,genre)VALUES('Attack on Titan','yash tailor','xyz','2004','1000','Horror');

INSERT INTO books(title,author,publisher,year_of_publication,price,genre)VALUES('Naruto','Uzumaki shibuya','abc','2008','1000','Kung-fu');

COMMIT;

Step 2 :- after COMMIT then make another insertion and perform a ROLLBACK

START TRANSACTION;

INSERT INTO books(title,author,publisher,year_of_publication,price,genre)VALUES('Black Clover','yami sukihero','manga','2015','2500','Magic');

ROLLBACK;

SCREENSHOT:-

Insert multiple records and then commit :-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)

```
START TRANSACTION;
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 8 (Query took 0.0011 seconds.)

```
INSERT INTO books(title,author,publisher,year_of_publication,price,genre)VALUES('Attack on Titan','yash tailor','xyz','2004','1000','Horror');
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 9 (Query took 0.0002 seconds.)

```
INSERT INTO books(title,author,publisher,year_of_publication,price,genre)VALUES('Naruto','Uzumaki shibuya','abc','2008','1000','Kung-fu');
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0015 seconds.)

```
COMMIT;
```

[Edit inline] [Edit] [Create PHP code]

	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	439	Fiction
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	385	Dystopian
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	2016	299	Romance
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	495	Classic
Edit Copy Delete	7	The Matrix	Sam Duol	James Hopper	2020	500	Action
Edit Copy Delete	8	Attack on Titan	yash tailor	xyz	2004	1000	Horror
Edit Copy Delete	9	Naruto	Uzumaki shibuya	abc	2008	1000	Kung-fu

Check all *With selected:* Edit Copy Delete Export

Insert another one record and then ROLLBACK :-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0001 seconds.)

```
START TRANSACTION;
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted.
Inserted row id: 10 (Query took 0.0008 seconds.)

```
INSERT INTO books(title,author,publisher,year_of_publication,price,genre)VALUES('Black Clover','yami sukihero','manga','2015','2500','Magic');
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0020 seconds.)

```
ROLLBACK;
```

[Edit inline] [Edit] [Create PHP code]

	book_id	title	author	publisher	year_of_publication	price	genre
Edit Copy Delete	1	The Alchemist	Paulo Coelho	HarperCollins	1988	439	Fiction
Edit Copy Delete	3	1984	George Orwell	Secker & Warburg	1949	385	Dystopian
Edit Copy Delete	4	Pride and Prejudice	Jane Austen	T. Egerton	2016	299	Romance
Edit Copy Delete	5	The Great Gatsby	F. Scott Fitzgerald	Charles Scribner's Sons	1925	495	Classic
Edit Copy Delete	7	The Matrix	Sam Duol	James Hopper	2020	500	Action
Edit Copy Delete	8	Attack on Titan	yash tailor	xyz	2004	1000	Horror
Edit Copy Delete	9	Naruto	Uzumaki shibuya	abc	2008	1000	Kung-fu

Check all *With selected:* Edit Copy Delete Export

Lab-26 : Set a SAVEPOINT before making updates to the members table, perform some updates, and then roll back to the SAVEPOINT.

Query :-

START TRANSACTION;

SAVEPOINT sp1;

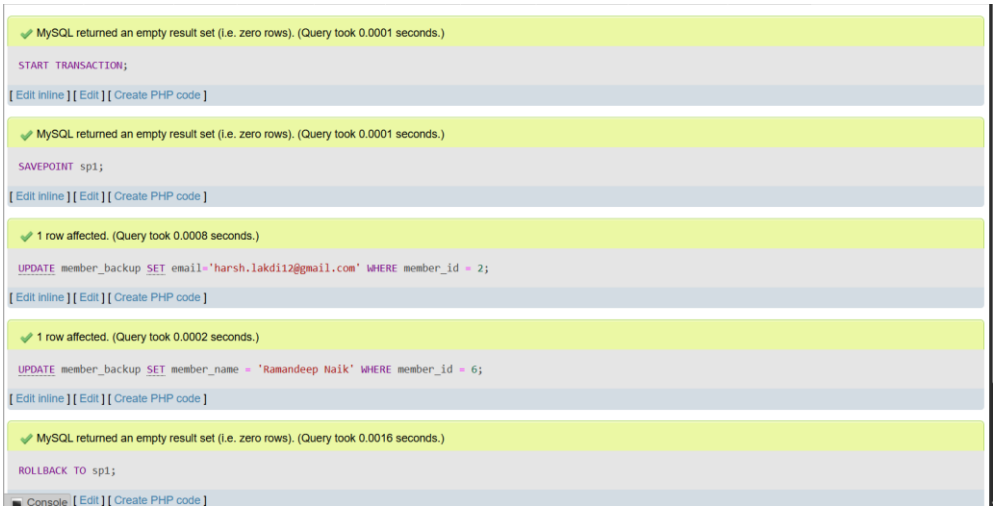
UPDATE member_backup SET email='harsh.lakdi12@gmail.com' WHERE member_id = 2;

UPDATE member_backup SET member_name = 'Ramandeep Naik' WHERE member_id = 6;

ROLLBACK TO sp1;

COMMIT;

SCREENSHOT:-



member_id	member_name	date_of_membership	email
1	Yash Tailor	2026-01-15	yash.tailor@gmail.com
2	Harsh Lakdiwala	2026-03-22	harsu.lakdiwala@gmail.com
3	Parth Gamit	2023-05-10	parth.gamit@gmail.com
4	Yash Gandhi	2023-07-18	yash.gandhi@gmail.com
5	Ronak Patel	2023-09-05	Ronak.patel@gmail.com
6	Rishi Naik	2020-04-14	rishi.naik@gmail.com
7	Aniket Patel	2026-04-15	ankit123@gmail.com

Lab-27 : Perform an INNER JOIN between books and authors tables to display the title of books and their respective authors' names.

Query :-

SELECT books.title, authors.first_name FROM books INNER JOIN authors ON books.author_id = authors.author_id;

SCREENSHOT:-

Extra options

title	first_name
The Alchemist	Jane
1984	George
Pride and Prejudice	Mark
The Great Gatsby	George
The Matrix	Jane
Attack on Titan	Mark
Naruto	Mark

Lab-28 : Use a FULL OUTER JOIN to retrieve all records from the books and authors tables, including those with no matching entries in the other table.

Query :-

SELECT books.title, authors.first_name FROM books LEFT JOIN authors ON books.author_id = authors.author_id;

UNION

SELECT books.title, authors.first_name FROM books RIGHT JOIN authors ON books.author_id = authors.author_id;

SCREENSHOT:-

Extra options

title	first_name
The Alchemist	Jane
1984	George
Pride and Prejudice	Mark
The Great Gatsby	George
The Matrix	Jane
Attack on Titan	Mark
Naruto	Mark

Lab-29 : Group books by genre and display the total number of books in each genre.

Query :-

SELECT genre, COUNT(*) AS total_books FROM books GROUP BY genre;

SCREENSHOT:-

genre	total_books
Action	1
Classic	1
Dystopian	1
Fiction	1
Horror	1
Kung-fu	1
Romance	1

Lab-30 : Group members by the year they joined and find the number of members who joined each year.

Query :-

SELECT date_of_membership AS join_date, COUNT(*) AS total_members FROM member_backup GROUP BY date_of_membership;

SCREENSHOT:-

Join_date	total_members
2020-04-14	1
2023-05-10	1
2023-07-18	1
2023-09-05	1
2026-01-15	1
2026-03-22	1
2026-04-15	1

Lab-31 : Write a stored procedure to retrieve all books by a particular author.

Query :-

CREATE PROCEDURE getbooksbyauthor()

SELECT *FROM book1 WHERE Author = 'A.P.J. Abdul Kalam';

CALL getbooksbyauthor();

SCREENSHOT:-

Extra options

BookID	Title	Author	Genre	Price
3	Wings of Fire	A.P.J. Abdul Kalam	Autobiography	250.00
4	Ignited Minds	A.P.J. Abdul Kalam	Motivational	220.00

Lab-32 : Write a stored procedure that takes book_id as an argument and returns the price of the book.

Query :-

```
DELIMITER //

CREATE PROCEDURE getbookprice(IN p_BookID INT)
BEGIN
   SELECT Price
   FROM book1
   WHERE BookID = p_BookID;
END //

DELIMITER ;

CALL getbookprice(2);
```

SCREENSHOT:-

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0043 seconds.)

```
CREATE PROCEDURE getbookprice(IN p_BookID INT) BEGIN SELECT Price FROM book1 WHERE BookID = p_BookID; END;
```

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 0 (1 total, Query took 0.0006 seconds.)

```
CALL getbookprice(2);
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Price
199.99

Lab-33 : Create a view to show only the title, author, and price of books from the books table.

Query :-

CREATE VIEW getbooksdetails AS SELECT Title, Author, Price FROM book1;

Execute Query:- SELECT *FROM getbooksdetails;

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```
CREATE VIEW getbooksdetails AS SELECT Title, Author, Price FROM book1;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

	Title	Author	Price
Edit Copy Delete	The Alchemist	Paulo Coelho	299.99
Edit Copy Delete	Brida	Paulo Coelho	199.99
Edit Copy Delete	Wings of Fire	A.P.J. Abdul Kalam	250.00
Edit Copy Delete	Ignited Minds	A.P.J. Abdul Kalam	220.00
Edit Copy Delete	The Monk Who Sold His Ferrari	Robin Sharma	275.50

Check all *With selected:* Edit Copy Delete Export

Lab-34 : Create a view to display members who joined before 2020.

Query :-

CREATE VIEW showmembers AS SELECT *FROM members WHERE join_date < '2020-01-01';

Execute Query:- SELECT *FROM showmembers;

SCREENSHOT:-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
CREATE VIEW showmembers AS SELECT *FROM members WHERE join_date < '2020-01-01';
```

[Edit inline] [Edit] [Create PHP code]

Extra options

	member_id	name	join_date
Edit Copy Delete	2	Raj Patel	2019-11-20
Edit Copy Delete	4	Parth Gamit	2017-08-25

Check all *With selected:* Edit Copy Delete Export

Lab-35 : Create a trigger to automatically update the last_modified timestamp of the books table whenever a record is updated.

Query :-

Step 1 :- first add the column last_modified in book1 table

ALTER TABLE book1 ADD last_modified TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP;

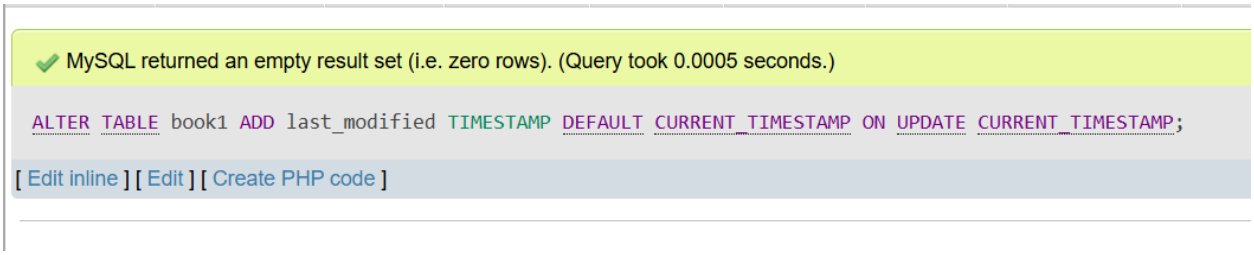
Step 2:- then create a trigger in book1 table

```
DELIMITER //


CREATE TRIGGER update_books_last_modified

BEFORE UPDATE ON book1

FOR EACH ROW

BEGIN

        SET NEW.last_modified = CURRENT_TIMESTAMP;

END //


DELIMITER ;
```

Step 3:- Update the record and check the timestamp

UPDATE book1 SET Price = 500 WHERE BookID = 3;

SCREENSHOT:-



MySQL returned an empty result set (i.e. zero rows). (Query took 0.0067 seconds.)

```
CREATE TRIGGER update_books_last_modified BEFORE UPDATE ON book1 FOR EACH ROW BEGIN SET NEW.last_modified = CURRENT_TIMESTAMP; END;
```

[Edit inline] [Edit] [Create PHP code]

☐	Edit	Copy	Delete	3	Wings of Fire	A.P.J. Abdul Kalam	Autobiography	500.00	2026-04-16 14:24:36

Lab-36 : Create a trigger that inserts a log entry into a log_changes table whenever a DELETE operation is performed on the books table.

Query :-

Step 1 :- creating a table for log_changes

CREATE TABLE log_changes(log_id int PRIMARY KEY AUTO_INCREMENT, BookID int, Title varchar(255), deleted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);

Step 2:- then create a trigger

```
DELIMITER //


CREATE TRIGGER after_delete_books

AFTER DELETE ON book1

FOR EACH ROW

BEGIN

        INSERT INTO log_changes(BookID, Title, deleted_at)VALUES(OLD.BookID, OLD.Title, NOW());

END //


DELIMITER ;
```

Step 3:- Delete the record and check the timestamp on log_changes table

DELETE FROM book1 WHERE BookID = 5;

Step 4:- retireve the data from the log_chnages table

SELECT *FROM log_chnages;

SCREENSHOT:-

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	log_id	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	BookID	int(11)			Yes	*NULL*			Change Drop More
3	Title	varchar(255)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
4	deleted_at	timestamp			No	current_timestamp()			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Spatial Fulltext

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0084 seconds.)

```
CREATE TRIGGER after_delete_books AFTER DELETE ON book1 FOR EACH ROW BEGIN INSERT INTO log_changes(BookID, Title, deleted_at)VALUES(OLD.BookID, OLD.Title, NOW()); END;
```

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
SELECT *FROM log_changes;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	log_id	BookID	Title	deleted_at
Edit Copy Delete	1	5	The Monk Who Sold His Ferrari	2026-04-16 14:44:20

Check all *With selected:* Edit Copy Delete Export

Lab-37 : Write a PL/SQL block to insert a new book into the books table and display a confirmation message.

Query :-

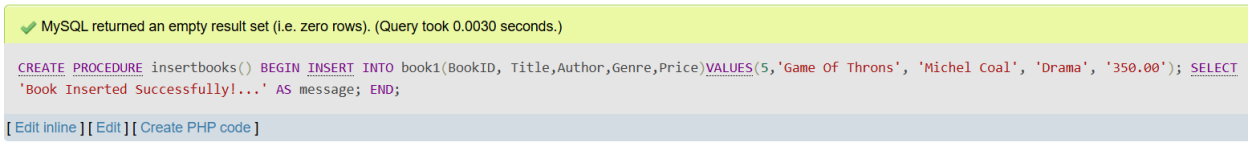
DELIMITER //

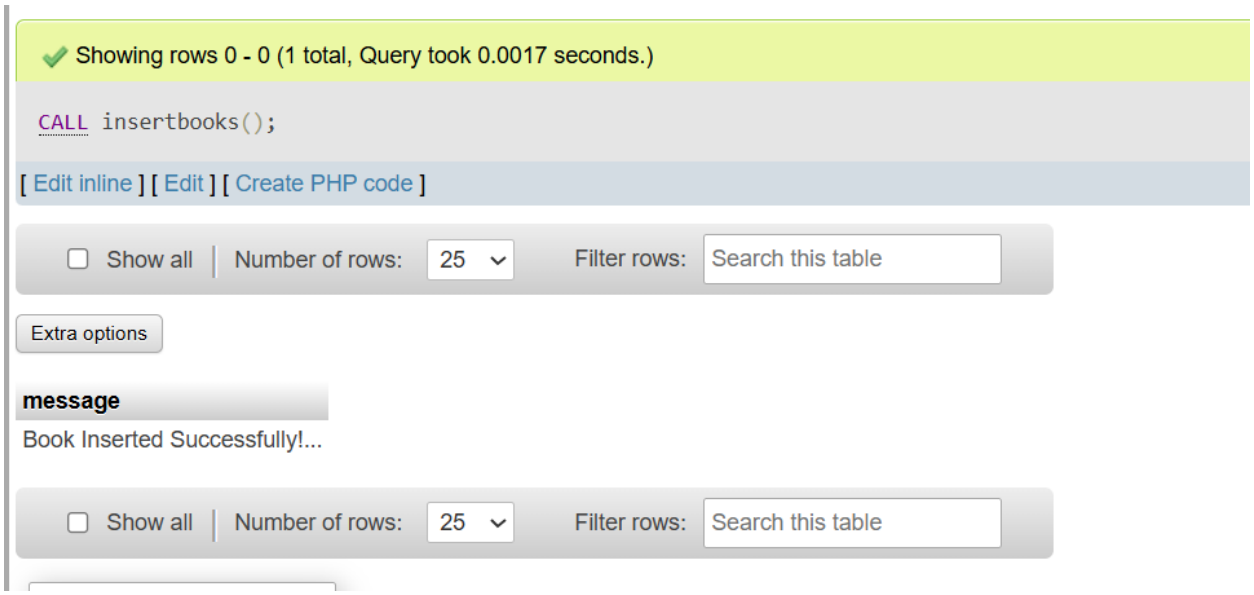
CREATE PROCEDURE insertbooks()

```
BEGIN
        INSERT INTO book1(BookID, Title,Author,Genre,Price)VALUES(5,'Game Of Throns',
'Michel Coal', 'Drama', '350.00');

  SELECT 'Book Inserted Successfully!...' AS message;
END //

DELIMITER ;
```

Execute Procedure :- CALL insertbooks();

SCREENSHOT:-





☐	Edit	Copy	Delete	4	Ignited Minds	A.P.J. Abdul Kalam	Motivational	220.00	2026-04-16 14:18:41
☐	Edit	Copy	Delete	5	Game Of Throns	Michel Coal	Drama	350.00	2026-04-16 14:54:54

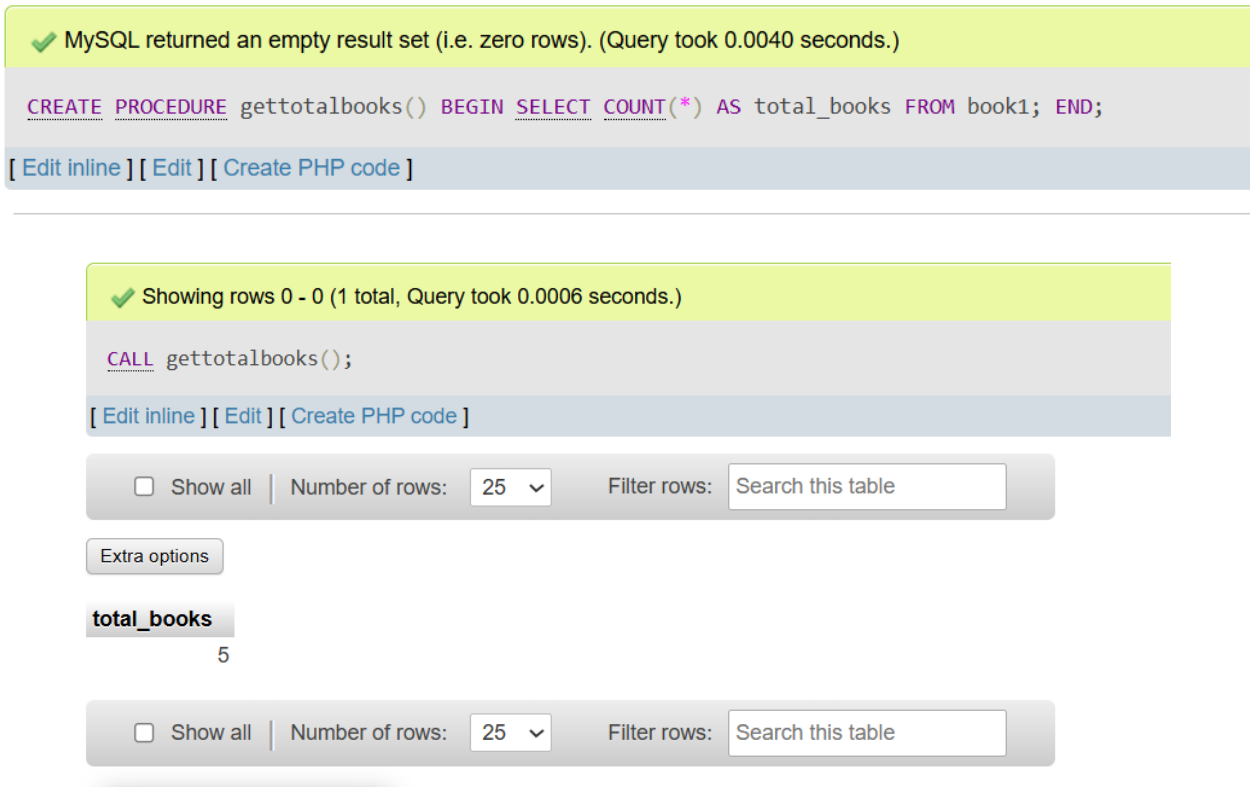
Lab-38 : Write a PL/SQL block to display the total number of books in the books table.

Query :-

```
DELIMITER //

CREATE PROCEDURE gettotalbooks()
BEGIN
        SELECT COUNT(*) AS total_books FROM book1;
END //

DELIMITER ;
```

Execute Procedure :- CALL gettotalbooks();

SCREENSHOT :-



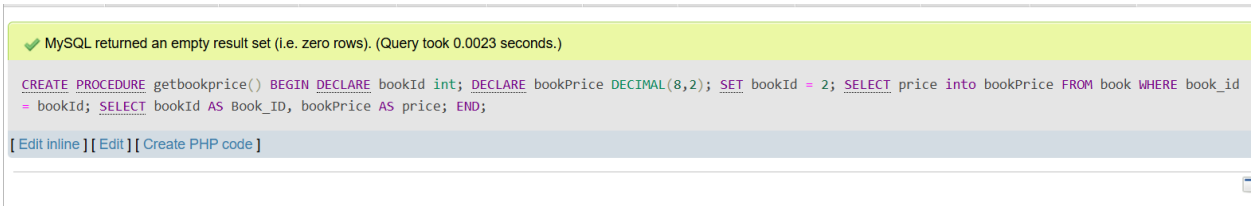
Lab-39 : Write a PL/SQL block to declare variables for book_id and price, assign values, and display the results.

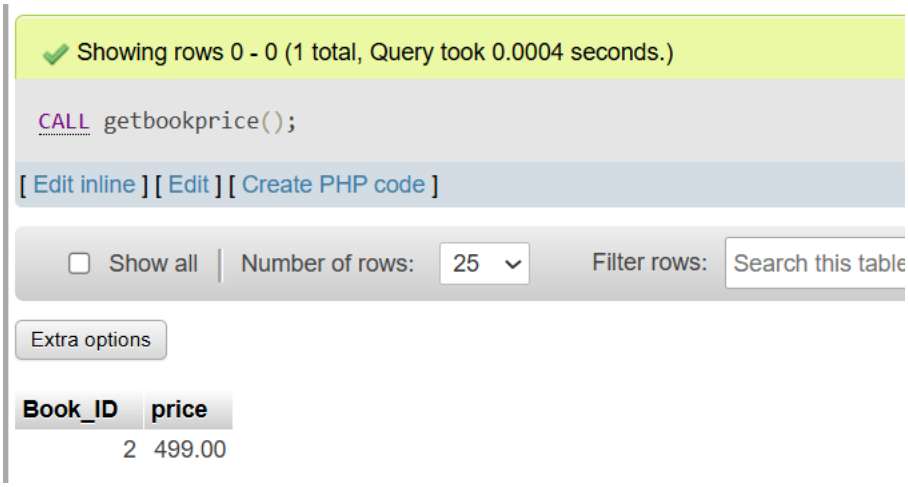
Query :-

```
DELIMITER //

CREATE PROCEDURE getbookprice()

BEGIN

   DECLARE bookId int;

   DECLARE bookPrice DECIMAL(8,2);


   SET bookId = 2;

   SELECT price into bookPrice

   FROM book

   WHERE book_id = bookId;


   SELECT bookId AS Book_ID, bookPrice AS price;

END //


DELIMITER ;
```

Execute Procedure :- CALL getbookprice();

SCREENSHOT :-





Lab-40 : Write a PL/SQL block using constants and perform arithmetic operations on book prices.

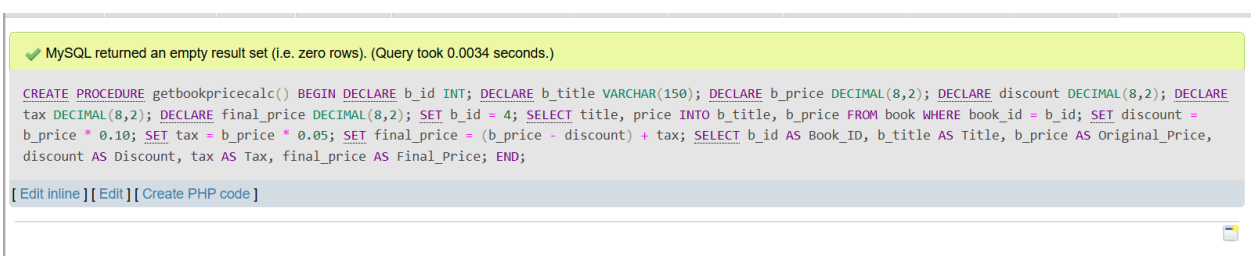
Query :-

```
DELIMITER //


CREATE PROCEDURE getbookpricecalc()

BEGIN

   DECLARE b_id INT;

   DECLARE b_title VARCHAR(150);

   DECLARE b_price DECIMAL(8,2);


   DECLARE discount DECIMAL(8,2);

   DECLARE tax DECIMAL(8,2);

   DECLARE final_price DECIMAL(8,2);


   SET b_id = 4;
```

```
    SELECT title, price
    INTO b_title, b_price
    FROM book
    WHERE book_id = b_id;

    SET discount = b_price * 0.10;
    SET tax = b_price * 0.05;
    SET final_price = (b_price - discount) + tax;

    SELECT
        b_id AS Book_ID,
        b_title AS Title,
        b_price AS Original_Price,
        discount AS Discount,
        tax AS Tax,
        final_price AS Final_Price;
END //

DELIMITER ;
```

Execute Procedure :- CALL getbookpricecalc();

SCREENSHOT :-



Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
CALL getbookpricecalc();
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Book_ID	Title	Original_Price	Discount	Tax	Final_Price
4	Think and Grow Rich	280.00	28.00	14.00	266.00

Show all | Number of rows: 25 Filter rows: Search this table

Lab-41 : Write a PL/SQL block using constants and perform arithmetic operations on book prices.

Query :-

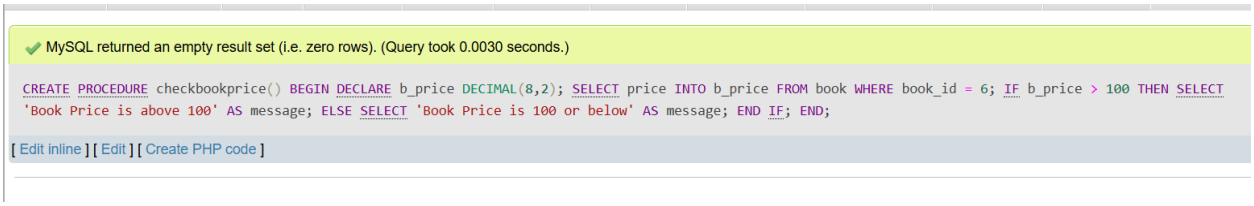
```
DELIMITER //

CREATE PROCEDURE checkbookprice()

BEGIN

        DECLARE b_price DECIMAL(8,2);


   SELECT price INTO b_price

   FROM book

   WHERE book_id = 6;


   IF b_price > 100 THEN

        SELECT 'Book Price is above 100' AS message;

   ELSE

        SELECT 'Book Price is 100 or below' AS message;

   END IF;
```

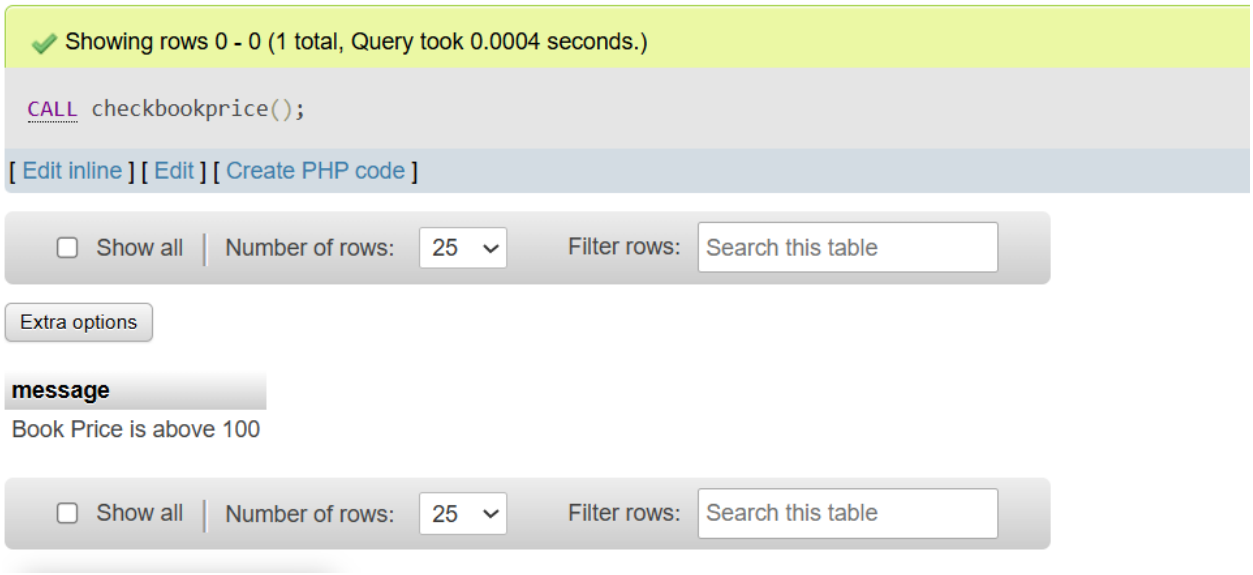
END //

DELIMITER ;

Execute Procedure :- CALL checkbookprice();

SCREENSHOT :-





Lab-42 : Use a FOR LOOP in PL/SQL to display the details of all books one by one.

Query :-

DELIMITER //

CREATE PROCEDURE displaybooks()

BEGIN

DECLARE isDone int DEFAULT 0;

```
        DECLARE b_id int;
    DECLARE b_title varchar(150);
    DECLARE b_price DECIMAL(8,2);

    DECLARE book_cursor CURSOR FOR
        SELECT book_id, title, price FROM book;
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET isDone = 1;

    OPEN book_cursor;

    read_Loop:LOOP
        FETCH book_cursor INTO b_id, b_title, b_price;

      IF isDone = 1 THEN
        LEAVE read_Loop;
      END IF;

      SELECT
        b_id AS Book_ID,
       b_title AS Title,
       b_price AS Price;
     END LOOP;

     CLOSE book_cursor;
END //
```

DELIMITER ;

Execute Procedure :- CALL displaybooks();

SCREENSHOT :-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0049 seconds.)

```
CREATE PROCEDURE displaybooks() BEGIN DECLARE isDone int DEFAULT 0; DECLARE b_id int; DECLARE b_title varchar(150); DECLARE b_price DECIMAL(8,2); DECLARE book_cursor CURSOR FOR SELECT book_id, title, price FROM book; DECLARE CONTINUE HANDLER FOR NOT FOUND SET isDone = 1; OPEN book_cursor; read_Loop:LOOP FETCH book_cursor INTO b_id, b_title, b_price; IF isDone = 1 THEN LEAVE read_Loop; END IF; SELECT b_id AS Book_ID, b_title AS Title, b_price AS Price; END LOOP; CLOSE book_cursor; END;
```

[Edit inline] [Edit] [Create PHP code]

Error

Static analysis:

1 errors were found during analysis.

1. Missing expression. (near "ON" at position 25)

SQL query: Copy Edit

SET FOREIGN_KEY_CHECKS = ON;

MySQL said:

#2014 - Commands out of sync; you can't run this command now

Lab-43 : Use a FOR LOOP in PL/SQL to display the details of all books one by one.

Query :-

START TRANSACTION;

INSERT INTO member(member_id,name,city)VALUES(6,'Sahil Patel','Surat');

SAVEPOINT sp1;

UPDATE member SET city = 'Indore' WHERE member_id = 6;

ROLLBACK TO sp1;

COMMIT;

SCREENSHOT :-

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0001 seconds.)

```
START TRANSACTION;
```

[Edit inline] [Edit] [Create PHP code]

1 row inserted. (Query took 0.0004 seconds.)

```
INSERT INTO member(member_id,name,city)VALUES(6,'Sahil Patel','Surat');
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0001 seconds.)

```
SAVEPOINT sp1;
```

[Edit inline] [Edit] [Create PHP code]

1 row affected. (Query took 0.0004 seconds.)

```
UPDATE member SET city = 'Indore' WHERE member_id = 6;
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
ROLLBACK TO sp1;
```

Console [Edit] [Create PHP code]

				member_id	name	city
Edit	Copy	Delete		1	Yash Tailor	Udaipur
Edit	Copy	Delete		2	Parth Gamit	Jaipur
Edit	Copy	Delete		3	Yash Gandhi	Ajmer
Edit	Copy	Delete		4	Harsh Lakdiwala	Delhi
Edit	Copy	Delete		5	Raj Patel	Mumbai
Edit	Copy	Delete		6	Sahil Patel	Surat

Check all *With selected:* Edit Copy Delete Export

Lab-44 : Use COMMIT after successfully inserting multiple books into the books table, then use ROLLBACK to undo a set of changes made after a savepoint.

Query :-

START TRANSACTION;

INSERT INTO book(book_id,title,price)VALUES(8,'Wisdom',420.00);

INSERT INTO book(book_id,title,price)VALUES(9,'How to make money',280.00);

SAVEPOINT sp1;

UPDATE book SET price = 800.00 WHERE book_id = 8;

ROLLBACK TO sp1;

COMMIT;

SCREENSHOT :-

book_id	title	price
1	The Alchemist	299.00
2	Atomic Habits	499.00
3	Rich Dad Poor Dad	350.00
4	Think and Grow Rich	280.00
5	The Power of Now	420.00
6	Ikigai	390.00
7	Deep Work	450.00
8	Wisdom	420.00
9	How to make money	280.00

Check all *With selected:* Edit Copy Delete Export